Review

Object State Maintenance in 3D Scene Graphs for Robotic Manipulation

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Abstract

Robots manipulating everyday environments need to remember more than object locations. Opening a drawer, moving a container, or storing an item changes object configurations, visibility, and relations, often without revealing the complete resulting state. This critical review examines how three-dimensional scene graphs represent and maintain such information. A purposive search of public papers and projects, updated on 19 October 2026, connects open-vocabulary mapping, dynamic scene reconstruction, interactive exploration, and articulated-object modeling. We organize the evidence around five maintenance operations: association, state estimation, invalidation, dependency repair, and verification. The comparison distinguishes geometric updates from discrete state changes and separates a system's representational capacity from the behavior actually evaluated. Existing methods provide complementary evidence for local map revision, interaction tracking, discovery of concealed objects, and articulation estimation; these results do not establish interchangeable solutions to complete state maintenance. We develop a state maintenance contract linking each graph assertion to its identity, temporal scope, evidential status, and downstream dependencies. A worked conceptual example and an evaluation checklist expose failures obscured by final task success or static graph accuracy. The resulting framework supports more precise system selection and motivates joint assessment of physical state, graph consistency, and recovery under partial observability.

Keywords: 3D scene graphs; robotic manipulation; object state; semantic mapping; partial observability; open-vocabulary perception; articulation; dynamic environments

1. Introduction

A household robot asked to retrieve a mug from a drawer must reason about a changing environment. The mug may be visible before storage, hidden afterward, and accessible only after another action. The drawer's location alone does not specify its opening configuration. Its opening configuration alone does not establish that the mug remains inside. These distinctions matter when an apparently sensible plan relies on a graph assertion that has become outdated or was never directly observed.

Three-dimensional (3D) scene graphs connect geometric entities with semantic descriptions and relations. Their appeal for manipulation lies in exposing information that can be queried by planners while retaining links to physical space. Early dynamic scene graphs connected objects, places, and humans, while later systems developed online construction and optimization [1, 2]. Open-vocabulary representations further enabled language queries beyond a fixed inventory of object labels [3, 4, 5]. These advances make scene graphs useful interfaces between perception and action. They also increase the number of assertions that a robot may retain and subsequently trust.

The relevant scientific question is therefore narrower than whether a scene graph can describe a scene. What evidence permits a particular assertion about an object to remain valid after interaction? A change in image appearance can indicate object movement, camera movement, articulation, or occlusion. A failed detection can indicate absence or insufficient visibility. An executed command can describe an intended transition without establishing its physical outcome. Treating these alternatives identically can produce a graph that remains linguistically coherent while losing its connection to the environment.

Recent broad reviews already organize the substantial literature on three-dimensional scene graphs. Catalano et al. analyze graph design, construction, and interaction across robotic applications [6]. Rotondi

et al. unify graph formulations and examine construction, applications, evaluation, and open challenges [7]. Both discuss dynamics and the mismatch between representations and downstream use. A further general taxonomy would duplicate much of that work. Our focus is the maintenance of manipulation-relevant object assertions across observed and unobserved changes.

This review makes three contributions within that scope. First, it separates identity, geometric configuration, discrete attributes, relations, and evidential status, rather than placing all changes under a single dynamic label. Second, it compares methods through five maintenance operations and distinguishes encoded information from evaluated behavior. Third, it develops a reporting contract and a set of diagnostic evaluation tasks. These are analytical products of the review, not a new mapping algorithm or an experimentally validated benchmark. Table 1 makes the relationship to adjacent reviews explicit.

Table 1. Positioning against adjacent reviews. The narrower scope below is this review's analytical choice, not evidence that broader reviews omit dynamics or manipulation.

Review	Organizing scope	Relationship to this review
Catalano et al. [6]	Graph design, construction, and interaction across robotics	Provides lifecycle context; we isolate validity of object assertions across manipulation events.
Rotondi et al. [7]	Graph definitions, pipelines, applications, evaluation, and challenges	Provides broad critical context; we connect identity, invalidation, repair, and verification at event level.
This review	Manipulation-relevant state maintenance under partial observability	Develops a reporting contract, bounded mechanism comparison, and proposed diagnostic evaluations; no new robot experiments.

The central argument is that update quality depends on what a graph retracts and verifies as well as what it adds. Local geometric revision, interaction histories, functional relations, and articulation models each address part of this problem. Their value depends on the observations available and the decision that the graph must support. We therefore compare information commitments and failure boundaries rather than combining incompatible task success percentages into a ranking.

2. Scope and Evidence Collection

2.1. Review design and research questions

We conduct a focused critical narrative review. This designation follows the distinction between review purposes and methods described by Grant and Booth [8]. The objective is a defensible synthesis of mechanisms and evidence boundaries, rather than an estimate of publication prevalence or a pooled treatment effect. The discussion is informed by narrative-review quality principles concerning explicit aims, search description, referencing, and scientific reasoning [9]. We do not claim exhaustive coverage or present a registered systematic review. No Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) screening flow is reported.

Three questions structure the extraction. RQ1 asks which object variables and relations a representation makes explicit. RQ2 asks what triggers revision, what evidence supports it, and which dependent assertions require reconsideration. RQ3 asks which evaluations distinguish a correct maintained graph from a successful final action. These questions determine the comparison fields and prevent a capability mentioned in a demonstration from being treated as a broadly established property.

2.2. Search and eligibility

The search was performed and updated on 19 October 2026. Public web searches targeted arXiv, publisher and conference records, author project pages, and public repositories. Query families combined scene graph terminology with dynamic updating, object state, manipulation, articulation, containment, and uncertainty. Named searches for DovSG, RoboEXP, MoMa-LLM, Khronos, DynaMem, Lost & Found, and articulated scene graphs provided starting points. References and related-work discussions in the two

recent broad reviews supplied additional discovery leads. The supplementary search log preserves the actual query strings, source URLs, access limitations, and verification actions.

Eligible core works describe an explicit three-dimensional graph used to construct, update, or query manipulation-relevant object information. We include adjacent mapping and memory systems when they clarify a maintenance mechanism that an explicit graph could use. DynaMem is such a boundary case: its dynamic spatial memory helps isolate geometric invalidation from graph-specific relational reasoning. Benchmarks are included when their observations or annotations could test identity, configuration, relations, or task outcomes. General language modeling, image-only scene graph generation, and policy learning are outside the core unless they support a specific comparison.

The main emphasis is work available from 2019 to the search date, with earlier methodological foundations included selectively. Public preprints are retained because several directly relevant articulated and uncertainty-aware approaches are recent. Their status is identified in the bibliography and evidence table. A later journal or conference record is preferred when its metadata can be verified. A project page establishes the existence of public materials; it does not establish that the code runs or reproduces reported results.

2.3. Extraction and limits of inference

For each retained source, the evidence table records its identifier, version or publication record, inspected material, claim locator, supported statement, and qualification. Method sections and available full texts were inspected for the central comparisons. Some contextual benchmark claims are supported only at abstract or documentation level, which is recorded explicitly. Full-text availability is kept separate from the extent of reading. Citation metadata and claim support are checked separately because a correct identifier does not ensure a correct interpretation.

The search is purposive and relevance-driven. Search-engine indexing, English-language emphasis, accessible full texts, and named starting systems can bias coverage. No database-wide retrieval denominator, duplicate-screening count, or independent reviewer agreement is claimed. These constraints limit statements about how common a design is across the entire field. They do not prevent a traceable comparison of the selected mechanisms. We extract no new robot measurements and perform no quantitative meta-analysis.

3. What Must Be Maintained

3.1. Identity and geometry

A graph node should distinguish an enduring object instance from its current observation. Let i denote an instance identifier and $T_i(t)$ its estimated rigid transformation in a specified reference frame. A new mask or point cluster is evidence about that instance, not automatically a new object. Conversely, semantic similarity between two mugs does not establish shared identity. Changing the reference frame after localization correction is also different from moving the physical object.

The re-localization problem in 3RScan makes this distinction concrete: object instances are matched across scans of changing environments [10]. SceneGraphFusion incrementally predicts graph information from color-and-depth (RGB-D) sequences, addressing the accumulation of object and relation evidence [11]. Neither multi-view fusion nor successful re-localization should, by itself, be read as evidence of arbitrary long-term manipulation-state maintenance. The review treats them as component capabilities with different observation assumptions.

3.2. Articulation and discrete state

Articulated state requires a configuration variable $q_i(t)$, an articulation model, and a reference configuration. For a rigid part attached through a modeled joint, its pose can be written schematically as

$$T_{part}(t) = T_{parent}(t)\, F(q_i(t)) \quad (1)$$

Here F is the joint transformation, not a learned transition model proposed by this review. The equation separates motion of the parent from motion within the articulated assembly. In a revolute joint q is an angle; in a prismatic joint it is a displacement. A binary label such as open is a task-dependent interpretation of configuration, not a substitute for the underlying geometry.

That interpretation requires an operational definition. A drawer can be open enough for visual inspection but insufficiently open for a gripper carrying a mug. A door can be ajar without permitting passage. Consequently, a published open/closed estimate should be interpreted with its threshold and intended action. More detailed kinematics can support trajectory generation, but estimation of a joint axis does not establish that a future actuation will succeed.

3.3. Relations and observability

Manipulation depends on relations with different physical meanings. Part-of links describe assembly structure; inside links describe containment; on links may represent geometric placement or physical support. Close-to links depend on distance conventions. Functional links can associate an interactive element with an object or an action. A relation name alone does not specify its direction, geometric test, causal meaning, or temporal validity. Table 2 separates these obligations.

Table 2. State variables and information obligations. These are analytical categories. Not every application needs every variable, and the examples are illustrative.

Information	Example	Maintenance obligation
Instance identity	Mug 17 across observations	Distinguish a moved instance from another similar instance or a duplicate node.
Rigid pose	World-frame object transform	Separate object movement from camera or map-frame correction; retain frame and time.
Articulation	Drawer displacement; door angle	Retain joint type, parent, reference configuration, limits if known, and configuration uncertainty.
Discrete attributes	Open; switched on; empty	Declare the sensing or inference route and the operational threshold; permit unknown values.
Object relations	Inside; on; part of; operates	Define direction and semantics; reconsider affected relations after endpoint changes.
Visibility and evidence	Occluded; inferred containment	Separate truth, visibility, provenance, and recency; missing detection is not automatic falsity.
Action accessibility	Open enough for this gripper	State task and embodiment assumptions; do not equate configuration with successful access.

Containment illustrates why geometry and knowledge should not be conflated. Closing a drawer can remove the mug from view without removing the mug or its remembered containment relation. Moving a drawer can change the mug's world pose if it moves with the drawer, but containment alone does not guarantee rigid co-motion. A mug behind a cabinet door may remain stationary when the door swings. The parent-child distinctions in articulated scene representations are therefore consequential, rather than cosmetic hierarchy choices [12, 13].

Partial observability also requires distinguishing an unknown assertion from a false one. A missing detection in an occluded region supports neither deletion nor confirmation. The decision-theoretic distinction between latent state and observation is established in partially observable planning [14]. A scene graph can expose an approximate memory of that state, but the presence of timestamps or confidence fields does not turn it into a calibrated belief distribution.

3.4. A state maintenance contract

We use state maintenance contract to denote a reporting checklist, not a software interface already shared by the reviewed systems. For each decision-relevant assertion, the checklist asks for an instance identifier, typed value or predicate, reference frame where relevant, observation time, asserted validity, evidential origin, and dependencies. The truth status of a Boolean predicate can be true, false, or unknown. Its origin can be observed, inferred, or predicted. These are separate fields: an inferred statement may be well supported, and an observed statement may be noisy.

A record can therefore take the conceptual form

$$r = (subject, predicate, value, truth, origin, t_{observed}, validity, source) \quad (2)$$

Quantitative values should carry uncertainty in a form appropriate to the estimator, or explicitly report that it is unavailable. The contract does not prescribe a probability for every field. Its purpose is to make unsupported certainty visible and to identify which planner queries require further sensing. The dependence of an assertion on older observations should survive graph serialization into a language-model prompt.

4. A Taxonomy of Maintenance Operations

4.1. Association preserves the subject of an update

Association determines which existing instance a new observation concerns. Spatial overlap, appearance, semantic similarity, motion continuity, and interaction context provide different evidence. Appearance is useful when an object moves, but similar instances can be confused. Geometric overlap is informative during small motion, but weaker after relocation or partial observation. Interaction tracking can bridge motion, provided the interacted object and the interaction interval are identified correctly.

These choices create distinct failure modes. Over-merging assigns observations of two objects to one node and transfers relations incorrectly. Over-splitting creates duplicate identities and leaves multiple plausible locations for one object. Identity swaps can preserve an accurate category-level map while corrupting instructions about a specific item. The appropriate error unit is consequently the instance and its history, rather than only the semantic class.

Open-vocabulary mapping does not remove these problems. ConceptGraphs combines object-centric geometric and semantic information in a queryable representation [4]. Open3DSG supports open-vocabulary objects and open-set relationships [15]. Those abilities broaden querying, while identity continuity remains a separate obligation. The recent hierarchical functional graph work explicitly addresses instance confusion and edge association across views, illustrating why relation inference depends on association quality [16].

4.2. Estimation changes values while retaining their meaning

Estimation updates a node's pose, joint configuration, attributes, or relational evidence. Accumulating observations can reduce noise when the underlying quantity is stable. The same averaging can blur a genuine transition when the object changes state. A maintenance system therefore needs a way to distinguish additional evidence for the same state from evidence that the state has changed. That decision can use motion, interaction events, or explicit temporal segmentation.

Articulated models impose useful structure. MoMa-SG estimates articulation from point trajectories and connects articulated objects with contained objects. Its configuration estimates are useful for subsequent manipulation, while the paper separately identifies semi-static rearrangement as future work [12]. ArtiSG extracts functional and kinematic information from human demonstrations, with a specialized acquisition interface that shapes its deployment assumptions [17]. These sources support a distinction between learning how an object moves and maintaining all later changes to that object.

Discrete attributes introduce an additional interpretation layer. An estimated joint displacement can support a thresholded open label; a successful skill return can support a tentative action outcome. Neither should overwrite direct evidence without an explicit reconciliation policy. A useful report states whether

state labels derive from sensing, simulator state, a classifier, or the controller's intended effect. Otherwise, two systems can report identical predicates while solving different inference problems.

4.3. Invalidation withdraws assertions that new evidence contradicts

Adding observations does not remove outdated geometry automatically. When a mug is moved, its earlier points can remain in a cumulative map. A graph constructed over that map can then contain a ghost object or an obsolete support relation. Invalidation is the operation that retracts, expires, or marks such assertions uncertain. It is logically different from detecting a new mug at the destination.

DovSG explicitly revises local memory, including obsolete voxel removal and subsequent graph updates [18]. DynaMem provides a useful adjacent mechanism: it uses new depth observations to remove outdated voxels that conflict with the visible free-space evidence [19]. These approaches illustrate observation-supported geometric invalidation. Their evidence should not be generalized to the deletion of a concealed object simply because it fails to appear in the current image.

Negative observations need a visibility model. If the camera could have observed the old location and the new depth measurement contradicts occupancy there, invalidation is more defensible. If the old location is hidden behind a closed door, the missing detection is uninformative about occupancy. This distinction suggests reporting both false deletion of persistent objects and retention of obsolete ones. A method can improve one error type while worsening the other.

4.4. Dependency repair restores compatible relations

A node update can invalidate assertions elsewhere in the graph. Moving a mug from a table into a drawer should reconsider its earlier on-table relation, its new containment, and any query that retrieves it through the table. Changing a room assignment can affect hierarchical retrieval without changing the object's local shape. Dependency repair denotes this propagation of reconsideration; it does not require every graph to use a symbolic inference engine.

The affected region is not necessarily a fixed spatial neighborhood. It can include articulation descendants, relations sharing an endpoint, or higher-level assertions derived from those relations. Lost & Found gives an explicit example by updating containment and proximity relations around a tracked interaction [20]. OpenObject-NAV highlights carrier relationships during navigation to moved instances [21]. These mechanisms motivate dependency-aware comparisons, although their different tasks do not establish a common optimal repair policy.

Physical constraints can help, but they require declared assumptions. A relation ontology might prohibit one object from having two exclusive immediate containers. That is appropriate only when the container granularity makes exclusivity meaningful. Nested containment and partial insertion complicate the rule. Similarly, carrying, containment, and support are not universally interchangeable. The contract therefore asks authors to state a constraint's intended domain rather than claiming unrestricted logical consistency.

4.5. Verification checks the result of action and revision

Verification asks whether the updated graph supports the next decision. A robot that issues open(drawer) has established a command event. It has not necessarily established an open drawer. Perception after execution, proprioception, contact information, or a verified controller condition can provide additional evidence. If those signals disagree, preserving uncertainty is more informative than recording the intended outcome as a fact.

Interactive exploration makes verification part of information acquisition. RoboEXP uses action-conditioned relations to reason about actions that reveal otherwise hidden objects [22]. MoMa-LLM integrates exploration and interaction with a dynamically updated graph for object search [23]. Their relevance is the coupling between action and subsequent observation. A complete maintenance evaluation would additionally separate the correctness of the post-action assertion from whether the eventual search task succeeds.

The five operations form a diagnostic decomposition, not a mandatory execution order. A system may jointly infer association and pose, or repeatedly alternate repair and verification. The decomposition is useful because it localizes failure. A wrong object identity requires a different remedy from an outdated relation, even when both cause the same grasp failure. Figure 1 shows the information flow and the distinct evidential boundaries.

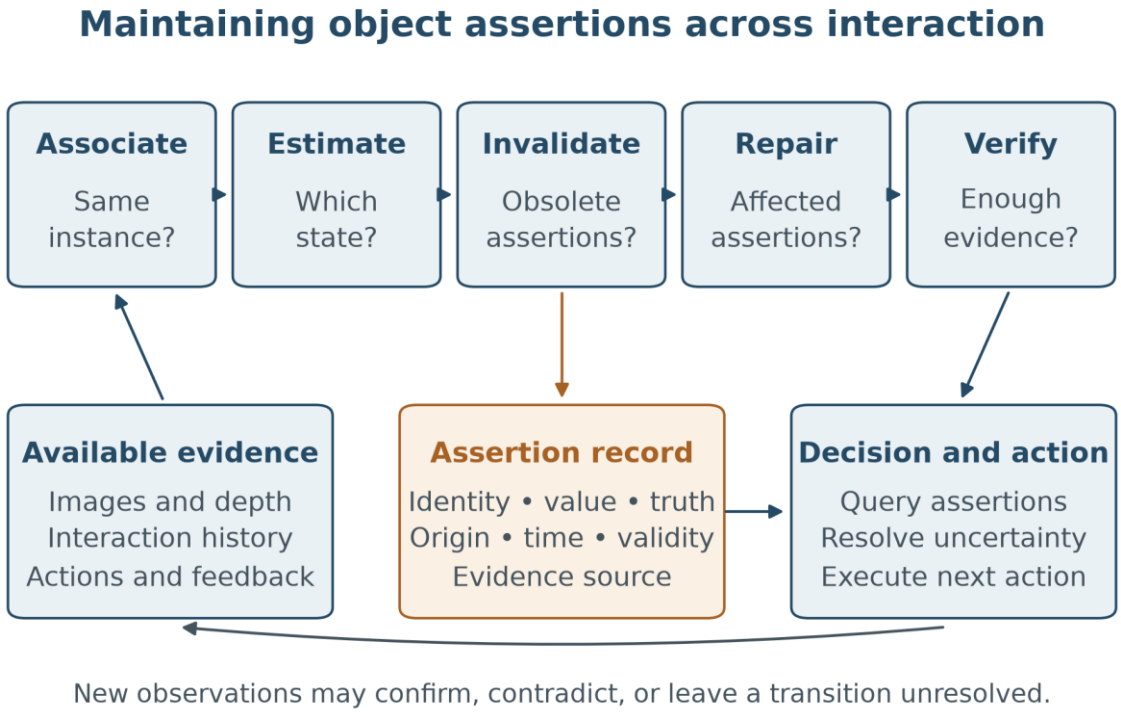


Figure 1. Five diagnostic operations and the evidence flow linking observations, maintained assertions, and decisions. The operations are an analytical decomposition and need not be implemented as separate modules or in a fixed order. This original schematic reports no measured performance.

5. Comparative Synthesis of Existing Approaches

5.1. Incremental mapping and dynamic reconstruction

Incremental graph construction and dynamic state maintenance should be compared separately. SceneGraphFusion updates graph predictions as RGB-D observations arrive. Hydra combines online graph construction with optimization [11, 2]. These are important foundations for a robot that cannot wait for a complete offline reconstruction. However, incremental processing alone says little about whether a graph retracts a no-longer-valid containment edge after manipulation.

Khronos explicitly models short-term dynamics and longer-term scene changes through complementary estimation processes [24]. Its contribution clarifies why dynamic mapping involves multiple timescales. A fast tracker can explain motion during observation, while later reconciliation can revise the persistent environment model. For manipulation, the remaining question is which configuration and relational assertions those estimates support. A metric-semantic temporal map and an action-conditioned graph may supply different answers without either being incorrectly designed.

Local revision is especially relevant when environmental changes are sparse relative to map size. DovSG connects localized memory adaptation with language-guided mobile manipulation. Its documented relationship vocabulary includes on, belong, and inside [18]. This is stronger evidence for selective geometric and relational revision than a purely static query demonstration. It does not establish a general continuous joint-state estimator. Table 3 records such differences without converting them into a universal capability score.

Table 3. Mechanism-oriented comparison of representative systems. Entries describe the inspected evidence, not exhaustive feature inventories or a performance ranking. Boundary statements concern the evidence available for this review; they do not assert that unreported capabilities are impossible. RGB-D denotes color and depth observations.

System and source	Information and revision mechanism	Evidence boundary for manipulation state
SceneGraphFusion [11]	Incremental object and relation prediction from RGB-D streams	Online accumulation is distinct from post-manipulation invalidation.
Hydra [2]	Online hierarchical graph construction and optimization	Map consistency does not by itself establish arbitrary object-attribute transitions.
ConceptGraphs / Open3DSG [4, 15]	Object-centric semantics and open-vocabulary relational queries	Queryable semantics and enduring instance identity are different requirements.
DovSG [18]	Localized geometric memory revision and scene-graph relation updates	Supports mobile manipulation after scene changes; not evidence of a general joint-state estimator.
Khronos [24]	Short-term tracking and longer-term metric-semantic reconciliation	Dynamic reconstruction and task-specific predicate verification have different targets.
DynaMem [19]	Dynamic spatial memory with depth-supported removal of stale voxels	Adjacent memory system, not an explicit relational scene-graph solution.
MoMa-LLM / RoboEXP [23, 22]	Exploration and interaction expose objects and update structured memory	Discovery depends on exploration and action capabilities; task success does not isolate assertion correctness.
Lost & Found [20]	Egocentric interaction tracking revises poses and containment/proximity edges	Requires observed interaction evidence; rigid-object tracking assumptions limit generalization.
OpenObject-NAV [21]	Carrier-relationship revision supports navigation to moved instances	Navigation evidence does not establish articulation estimation or complete action-state maintenance.
OpenFunGraph / hierarchical extension [25, 16]	Interactive-element detection and functional-edge inference or temporal association	Functional relations are not equivalent to continuous joint configuration or action postconditions.
ArtiSG / Pandora [17, 13]	Human interaction observations supply kinematic or object-container information	Demonstration acquisition, alignment, and partial observation shape the applicable scope.
MoMa-SG [12]	Point-track articulation estimation, containment structure, and configuration-aware manipulation	Prismatic/revolute and sensing assumptions apply; semi-static rearrangement remains a stated future direction.
ST-OVSG [26]	Temporal association and latency annotations support teleoperation planning	Latency-aware history is different from sensing an unobserved physical intervention.
Semantic uncertainty [27]	Object reliability is aggregated into hierarchical semantic assertions	Semantic reliability does not constitute a full action-transition belief model.

5.2. Interaction histories and concealed objects

Direct observation of a transition can provide information that a later rescan cannot recover. Lost & Found tracks interacted rigid objects from egocentric observations and maintains a transformable graph, including drawer containment [20]. The important consequence is historical: an object can remain queryable after storage even when its current appearance is unavailable. This conclusion depends on observing the relevant interaction and preserving the correct association.

Interactive exploration addresses a different visibility problem. The robot may not know that an object exists until it opens or moves an obstruction. An action-conditioned relation can identify the actions

required to obtain that observation [22]. Remembering that a mug was placed in a drawer and learning that a drawer contains a mug through exploration are complementary information routes. Their evaluation should distinguish persistence of known objects from discovery of previously unknown objects.

Interventions outside the sensor's view create a boundary shared by both routes. Historical containment remains a hypothesis if another agent can rearrange the scene unobserved. A graph can reasonably preserve the last supported location, but it should expose the observation time and the possibility of change. Resolving that ambiguity requires additional evidence or a stated persistence model. No graph structure alone supplies information about an unobserved event.

5.3. Functional relations and articulated configuration

Functional graph construction expands what counts as a manipulation-relevant entity. OpenFunGraph predicts objects, interactive elements, and functional relationships from posed RGB-D observations, and evaluates them using functional graph annotations [25]. Such a relation can connect a handle to the object it operates. This provides an affordance-relevant structure, while a joint model specifies the motion compatible with operating that structure. Functional labeling and kinematic estimation should therefore remain separate columns in comparisons.

Recent approaches connect these layers more directly. Pandora derives articulation information and object-container relations from human egocentric exploration; it reports limitations under partial observations and restricts its dynamic-object scope [13]. ArtiSG uses demonstrations to recover functional elements and kinematic priors [17]. MoMa-SG estimates semantic-kinematic graphs and evaluates articulated manipulation, with explicit assumptions concerning articulation types and sensing [12]. These advances substantially narrow the geometry-action gap without establishing unrestricted maintenance of arbitrary object attributes.

The comparison also reveals an acquisition trade-off. Passive observations impose fewer interaction requirements but can leave mechanisms ambiguous. Human demonstrations provide transition evidence but require collection and alignment. Autonomous exploration obtains task-specific observations but inherits the robot's manipulation limits. The appropriate choice depends on whether the graph is built once for deployment, updated during tasks, or continually revised by multiple agents. Papers should report this lifecycle because it determines what maintenance claims are justified.

5.4. Time, uncertainty, and task relevance

Different temporal representations answer different questions. A timestamp supports recency checks; a history supports retrospective queries; a transition model predicts state; a validity interval limits when an assertion is claimed. ST-OVSG explicitly addresses temporal information and communication latency in teleoperation [26]. Aion and Rheos instead model aggregate motion dynamics for navigation [28, 29]. These works demonstrate why temporal reasoning cannot be reduced to the presence of a time index.

Uncertainty also has multiple targets. A detector score concerns a perception module's output. Geometric covariance concerns a pose estimate under a model. Semantic uncertainty concerns the correctness or suitability of a label. Recent work on hierarchical semantic uncertainty propagates object reliability toward room-level assertions [27]. Its relevance is the explicit treatment of unequal evidence strength. This does not by itself characterize uncertainty over the outcomes of opening a drawer or over an unseen intervention.

Task relevance can reduce the information that needs to be retained or queried. HOV-SG supplies hierarchical language-grounded navigation, while Clio adapts graph granularity to a specified task set [30, 31]. SayPlan uses hierarchical graph structure for scalable task planning [32]. For maintenance, compression should preserve enough provenance and dependencies to recognize when a retained assertion requires revision. Whether particular compression policies achieve this remains a separate empirical question.

6. What Existing Evaluations Establish

6.1. Distinct evidence layers

Evaluation can target at least four different objects: perceptual primitives, graph assertions, temporal updates, and downstream actions. Good instance segmentation does not ensure a correct support relation. Accurate static graph prediction does not ensure that a relation is withdrawn after change. A successful task does not reveal whether the planner compensated for an incorrect graph by looking again. These are logical distinctions between evaluation targets, rather than claims that every published system neglects them.

The graph benchmark introduced with 3DSSG supports object and relation prediction in indoor reconstructions [33]. 3RScan supports cross-scan instance correspondence and re-localization in changed scenes [10]. Functional graph resources associated with OpenFunGraph and the later hierarchical functional work expand the evaluated entities and relations [25, 16]. Their annotations provide useful evidence, but an online manipulation-maintenance task requires additional event timing and observation information. Table 4 identifies what each resource contributes and what must be added for that use.

Table 4. Public evaluation resources and their relevance. The final column is this review's assessment of additions needed for the proposed use, not a criticism of each resource's original objective. No new evaluation on these resources is reported.

Resource	What it can inform	Needed for event-level maintenance evaluation
3RScan / RIO [10]	Instance correspondence and object re-localization across changed scans	Observation timing and action/intervention histories between snapshots.
3DSSG [33]	Object and relation prediction in indoor reconstructions	Time-indexed relation changes and visibility-conditioned update labels.
FunGraph3D / SceneFun3D extensions [25, 16]	Interactive elements, functional links, and hierarchical associations	Separate physical state transitions from multiview association; label stale edges.
Arti4D-Semantic [12]	Articulation estimates and parent-child object relations	Broader interventions and common event-level recovery protocols.
SAPIEN / ManiSkill2 [34, 35]	Controlled articulated and manipulation transitions	Fix robot observations; withhold privileged state from graph inference.
RLBench [36]	Visual manipulation tasks and multistage activities	Persistent instance and relation traces across actions and failures.
ALFRED / TEACh [37, 38]	Language-conditioned household state changes and interaction	A declared graph schema and assertion-level annotations alongside task labels.
BEHAVIOR-1K [39]	Long-horizon activities with physical and semantic object properties	A matched maintenance protocol with event truth separated from accessible observations.
Ego4D [40]	Human interaction and visual object-state-change observations	Persistent metric identities and robot-query alignment beyond local video labels.

6.2. Simulation and embodied task benchmarks

Interactive simulation offers repeatable transitions and access to physical state. SAPIEN supports part-based interactive environments, while ManiSkill2 provides a range of manipulation tasks and observation interfaces [34, 35]. RLBench offers visual manipulation tasks, including articulated and multistage activities [36]. These resources are useful for testing action-conditioned changes under controlled settings. Using privileged simulator state to build the graph, however, bypasses the perceptual inference problem that maintenance is intended to solve.

Language-conditioned benchmarks expose other aspects of state dependence. ALFRED includes compositional household tasks with non-reversible state changes, and TEACh adds interactive dialogue in

task execution [37, 38]. BEHAVIOR-1K provides long-horizon household activities and rich physical and semantic object properties [39]. These are opportunities to construct maintenance evaluations, not evidence that a common graph-maintenance protocol already exists. The tested observation regime and the source of success labels should be reported explicitly.

Egocentric video supplies observations of human-object interaction beyond a robot's own data stream. Ego4D includes object-state-change tasks within a wider egocentric benchmark collection [40]. Such data can inform state-transition perception. Connecting a local visual state change to a persistent metric instance, a containment history, and a robot's later query requires additional alignment and annotation. Evidence for one component should not be promoted to evidence for the entire chain.

6.3. Why reported success rates should not be pooled

The reviewed methods use different sensors, scene preparation, action repertoires, environmental interventions, and success criteria. Some estimate map structure from recorded observations; others execute exploration or manipulation. Articulation estimation, hidden-object retrieval, and language-guided pick-and-place do not share a common denominator. We therefore do not reproduce a league table of their headline percentages. Direct numerical comparison would require matched tasks and observation conditions.

Even within a shared task family, maintenance errors can be masked. A graph may hold an incorrect location until the robot approaches and redetects the target. The task succeeds, yet the stored assertion was unsuitable for decisions earlier in the episode. Conversely, a mechanically failed grasp may leave an accurate graph and still produce task failure. Separating these cases requires event-level observations and causal attribution of failure, not only an endpoint label.

7. An Evaluation Agenda for State Maintenance

7.1. Event-level records

We propose recording each episode as a sequence of state-changing or information-changing events. Each event should identify the involved instances, attempted action or external intervention, available observations, graph assertions before and after revision, and an independently supported outcome. Physical state labels and observation visibility labels should be distinct. The aim is to test whether the graph uses available evidence appropriately, without rewarding knowledge that the robot could not have acquired.

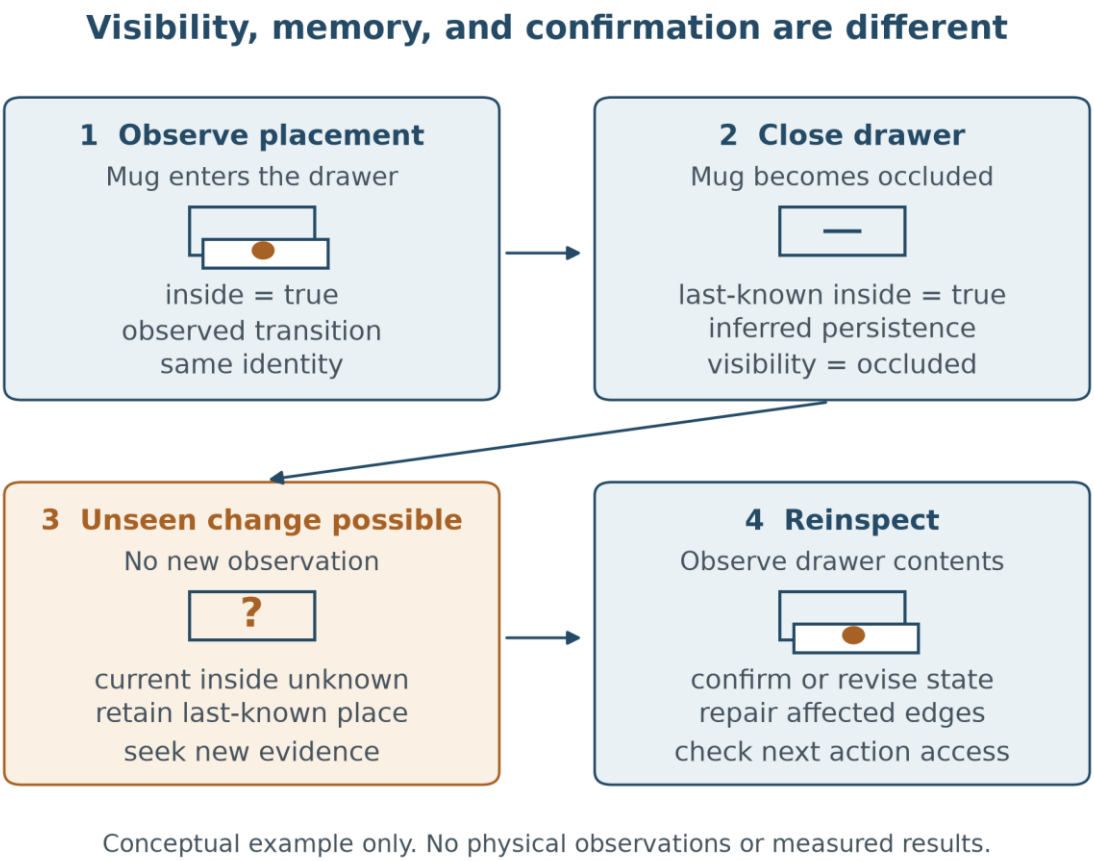


Figure 2. A conceptual drawer episode separating observed placement, remembered containment, possible unseen change, and reinspection. An unobserved intervention is a possibility, not an asserted event. The example illustrates information states and is not experimental data.

Figure 2 gives a conceptual drawer episode. The sequence distinguishes visible placement, remembered containment after closure, an unobserved possible intervention, and active reinspection. It is an explanatory example with no measured results. When the drawer closes, the mug becomes occluded; that event does not make containment false. When unobserved intervention is permitted, last-known containment and current confirmed containment cease to be equivalent. Reinspection can resolve the uncertainty if the relevant volume is actually observed.

Such episodes should include repeated instances, imperfect sensing, unsuccessful actions, and changes outside the robot's view. The evaluator should label an assertion unidentifiable when the permitted observations do not distinguish competing physical states. A method that abstains in this case should not be penalized as if it missed visible evidence. Conversely, a method that asserts certainty should not receive credit merely because its guess matches the hidden state.

7.2. Diagnostic metrics

Table 5 proposes metrics that complement task success. They are reporting recommendations, not quantities measured in this review. A false-deletion rate concerns existing instances removed or declared absent without sufficient support. A stale-assertion exposure rate concerns decision queries that receive an assertion contradicted by the event ground truth at query time. An identity-switch measure tracks whether successive observations are assigned to the same physical instance. Each metric should disclose its eligible event set and treatment of unknown labels.

Table 5. Proposed diagnostic reporting measures. All entries are recommendations and none is a measured result of this review. Denominators and unknown cases must be disclosed; ground truth should be independent of the graph under test.

Measure	Eligible unit and definition	Required qualification
Identity switches	Changes of assigned graph identity along an annotated physical-instance track	Report track fragmentation and occlusion regime separately.
False deletion	Persistent, evaluable instances removed or asserted absent without sufficient evidence, divided by eligible persistent instances	Separate deletion of the identity from invalidation of an old pose.
Stale assertion exposure	Outdated assertions returned at evaluable decision queries divided by evaluable queries	Also report unknown/abstention frequency, coverage, and task cost.
State error	Pose error, joint error, or discrete-predicate error on independently labeled observations	State units, joint reference, thresholds, and visible versus inferred status.
Relation violations	Counts of declared-domain constraint violations per evaluated graph snapshot	Publish constraint definitions; nested or partial containment may be valid.
Verification precision	Independently true postconditions among postconditions claimed verified	Report missed verification and coverage to prevent selective reporting.
Repair delay	Elapsed time from physical change and, separately, first informative observation to a correct assertion	Censor unresolved cases explicitly; do not discard them from delay summaries.
System cost	Sensing actions, update latency, memory, and task outcome per episode	Keep observations, controller, and planner matched when isolating maintenance.

One useful query-level definition is

$$S = N_{stale} \,/\, N_{evaluable\ queries} \quad (3)$$

where N_{stale} counts current-state queries answered with an outdated assertion. $N_{\text{evaluable_queries}}$ counts current-state queries whose relevant physical truth is independently known. Unknown or abstained answers should be reported separately, including their frequency and task consequences. Otherwise, indiscriminate abstention could artificially reduce S. This diagnostic measures exposure to stale information; it is not a generic accuracy score for the whole graph.

Repair delay should also distinguish physical events from first informative observations. The delay from an event to a correct graph measures end-to-end responsiveness. The delay from the first observation sufficient to identify that event measures processing and inference latency. A robot cannot be expected to update from evidence it has not received, but a deployed system still bears the cost of delayed sensing. Reporting both delays makes that trade-off visible.

7.3. Minimal controlled comparisons

A useful comparison would keep the perception backbone, observations, task planner, and action controller fixed while varying maintenance policy. Candidate policies include cumulative fusion, local geometric invalidation, interaction-guided revision, and revision with explicit dependency checks. Such a study could isolate which error categories each policy changes. It would need to report runtime and observation costs because frequent complete rescanning can improve state accuracy while consuming substantial task time.

Different oracle conditions would further localize failure. Verified instance associations could isolate state-update errors; verified action outcomes could isolate sensing failures; verified graph assertions could measure the remaining planning and control limitations. These are diagnostic interventions, not deployment assumptions. Their purpose is to avoid assigning all end-to-end failure to scene-graph reasoning when perception or mechanics is responsible.

Public code availability is helpful but insufficient for this comparison. A reproducible report should specify the actual repository revision, model checkpoints, prompts, thresholds, sensor calibration, and observation-to-action timing. For remote models, version and request timing can affect behavior. A release should include event traces and failed episodes as well as successful demonstrations. The present review provides a schema and reporting checklist to support such future studies; it does not claim to have executed them.

8. Discussion and Research Priorities

8.1. State maintenance as information management

Across the selected literature, the most useful distinction is between acquiring more information and maintaining the validity of existing information. Open-vocabulary features expand the queries that can be posed. Interaction tracking preserves transitions that may later be invisible. Articulation models constrain feasible configuration changes. None of these alone specifies when a planner should stop trusting an old assertion. The state maintenance contract makes that decision visible across otherwise different architectures.

This perspective complements the lifecycle and evaluation analyses of recent general reviews [6, 7]. It does not claim that object state, uncertainty, or dynamics were previously overlooked. The narrower contribution is a common set of questions for comparing their operational treatment. These questions connect identity, invalidation, dependent relations, and post-action verification to the same decision episode. They also make explicit when two papers use similar terminology for different information commitments.

8.2. Priorities that follow from the evidence

The first priority is to connect geometric updates with relational validity. Free-space evidence can remove outdated geometry, while interaction histories can preserve concealed objects. Combining these mechanisms requires rules that respect visibility and identity rather than indiscriminate deletion. The discriminating test is whether a system retracts an obsolete location while retaining a supported hidden-object hypothesis under the same sensing budget.

The second priority is to distinguish stable mechanism knowledge from transient configuration and task accessibility. A drawer's estimated joint axis can remain useful after its opening changes. A handle relation may remain useful even when the handle is occluded. A reachable or open-enough predicate depends additionally on embodiment and task. Evaluation should therefore separate mechanism error, configuration error, and action-feasibility error rather than compressing them into an articulated-object success label.

The third priority is to make uncertainty actionable. Storing a confidence score is useful only if the downstream system knows what uncertainty it describes and how to respond. An uncertain identity may call for a discriminative view; uncertain containment may call for opening a container; uncertain grasp execution may call for a postcondition check. These choices require different observations. Future evidence should establish whether such targeted verification reduces errors relative to uniform reobservation at comparable cost.

8.3. Limitations and alternative interpretations

This review's taxonomy is an interpretive synthesis. Its five operations are not mutually exclusive modules, and no experiment here shows that implementing all five improves performance. Some applications may reasonably use a smaller state model. A fixed-workcell pick-and-place task may require little history, whereas searching after human rearrangement may require explicit persistence and visibility. The contract should be scaled to the assertions that actually affect the robot's decisions.

Several apparent maintenance benefits may instead arise from stronger perception, better localization, simpler scene preparation, or repeated sensing. The selected papers do not provide a single controlled experiment that separates all these factors. This limits causal claims about graph structure. It also explains

why our recommendations emphasize matched observations and component-level oracle conditions. A richer graph is not inherently more useful if its added assertions are unreliable or unused.

Finally, the available evidence is uneven across object properties. Position, articulated configuration, containment, and interactive elements are better aligned with the selected systems than deformable shape, material state, liquids, or contact-force history. Simulation can represent some of these variables, but that does not establish reliable inference from deployment observations. Our conclusions concern the maintenance of explicitly represented manipulation state in the reviewed settings and should not be read as a complete account of embodied world modeling.

9. Conclusions

Maintaining object state in a three-dimensional scene graph requires more than repeatedly detecting objects. A useful update preserves instance identity, revises physical estimates, withdraws contradicted assertions, repairs affected relations, and checks the information needed for the next action. The reviewed literature provides concrete but complementary mechanisms for these operations. Local map revision, egocentric interaction histories, active exploration, functional graphs, and articulated models address different observability and deployment conditions.

The main contribution of this review is an evidence-aware comparison of those conditions through a state maintenance contract. Its worked example and proposed diagnostics expose distinctions that static graph accuracy and final task success cannot resolve alone. The contract and metrics remain recommendations requiring empirical validation. Their immediate value is to make claims about dynamic graphs more precise and to identify the observations, annotations, and controlled comparisons needed to test reliable manipulation-state maintenance.

Supplementary Materials

Supplementary File S1 contains the source evidence table, reference verification records, and search log. Supplementary File S2 contains the state maintenance reporting checklist and an illustrative event-record schema. The schema is a proposed documentation aid and contains no experimental observations.

Author Contributions

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Institutional Review Board Statement

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Informed Consent Statement

Not applicable.

Data Availability Statement

The source identifiers, search record, literature evidence table, and proposed reporting schema supporting this review are provided in the Supplementary Materials. No new experimental data were generated or analyzed. Public source materials remain available through the cited publishers, archives, and project repositories, subject to their respective access conditions and licenses.

Conflicts of Interest

The author declares no conflict of interest.

Abbreviations

3D, three-dimensional; RGB-D, color and depth; SLAM, simultaneous localization and mapping; LLM, large language model; VLM, vision-language model; RQ, research question.

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